

Technische Hochschule Ingolstadt
Fakultät Elektro- und Informationstechnik

Sensor Network Technologies and Sensor Fusion

Particle Filters

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Welcome to the lecture!

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AIMotion, AI and Sensor Data Fusion

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- **Please register for the lecture on Moodle!**
  - Link: https://moodle.thi.de/course/view.php?id=10580
  - Key: **SensNT+DF#WS25**
- **News on Moodle:**
  - Slides for the lecture (upcoming lectures marked as WIP and are subject to change!)
  - Formulary, Exercises (Solutions provided after discussion in the lecture)
  - Exercises are optional, one task / sheet per week, but are highly recommended.
    They can be handed in for correction.
  - I will *try* to publish exercises before Wednesday‘s lecture so that initial questions can be discussed on Thursday.
- **Exam: Written exam, 90 minutes**
- **Consultation hours: Please arrange via e-mail to bjoern.giesler@thi.de.**
- **Please also don‘t hesitate to send questions, lecture feedback, change requests, etc.!**

- **In this lecture:**
  - Quick recap of the Extended Kalman Filter
  - Derivation of Particle Filters
    - Basic Idea
    - (Sequential) Importance Sampling
    - Transitional Prior
    - Resampling
- **After that, we will revisit tracking briefly using Joint Probability Data Association Filters, and their combination JPDA-PF with Particle Filters.**
- **The rest of the lecture series will deal with**
  - Simultaneous Localization and Mapping, and
  - (DNN-based) early fusion approaches.

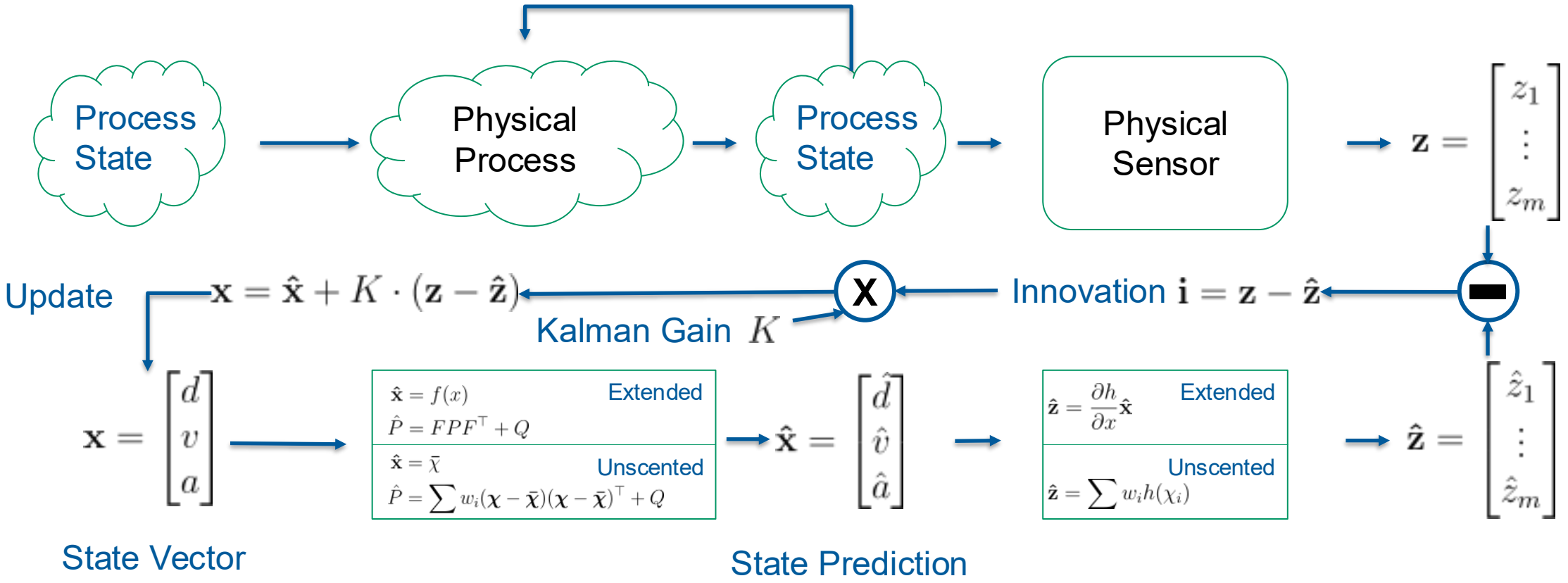
Sheet 5 – Solution

Let's talk exam!

- **I want to know what's in your brain! My priority is always**
  1. have you understood the concept?
  2. ...thoroughly enough so you could implement it practically?
  3. and last, are the numerical results you obtain correct?
- **No formulary, personal notes, tablets, laptops, AI assistants or personal humanoid messenger or telepresence robots allowed.**
- **Pocket calculators are a necessity.**
- **There will likely be some hand calculation with the help of pocket calculators.**
- **No coding of course, but questions about algorithms in pseudo code are possible.**
- **Long answer questions possible. Multiple choice questions possible. Formula questions possible.**
- **There will be more exercise sheets (at least two, maybe three).**
- **Basically everything we discussed in the lecture is relevant. Do not rely on only the exercise sheets or programming exercises to prepare.**

Particle Filters

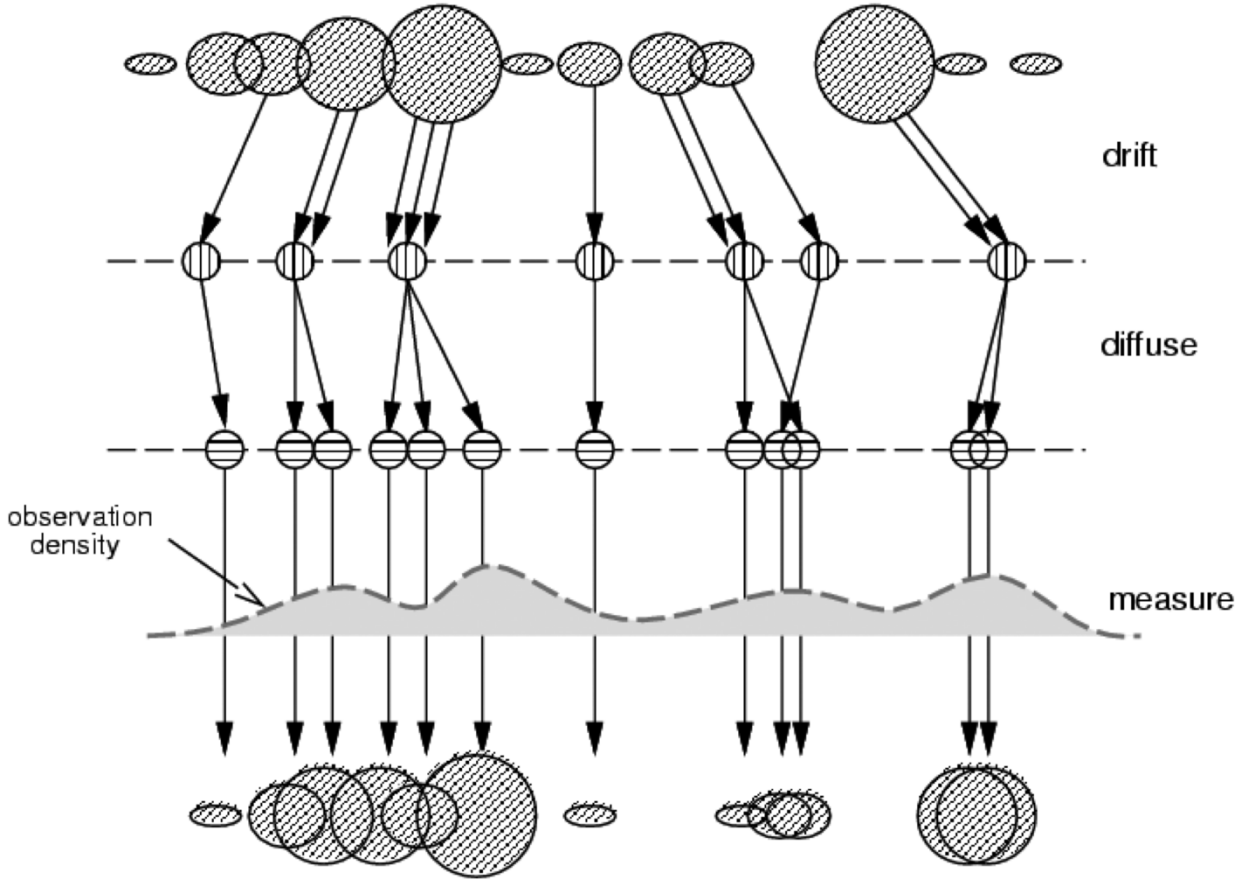
Approximating State Transition Functions





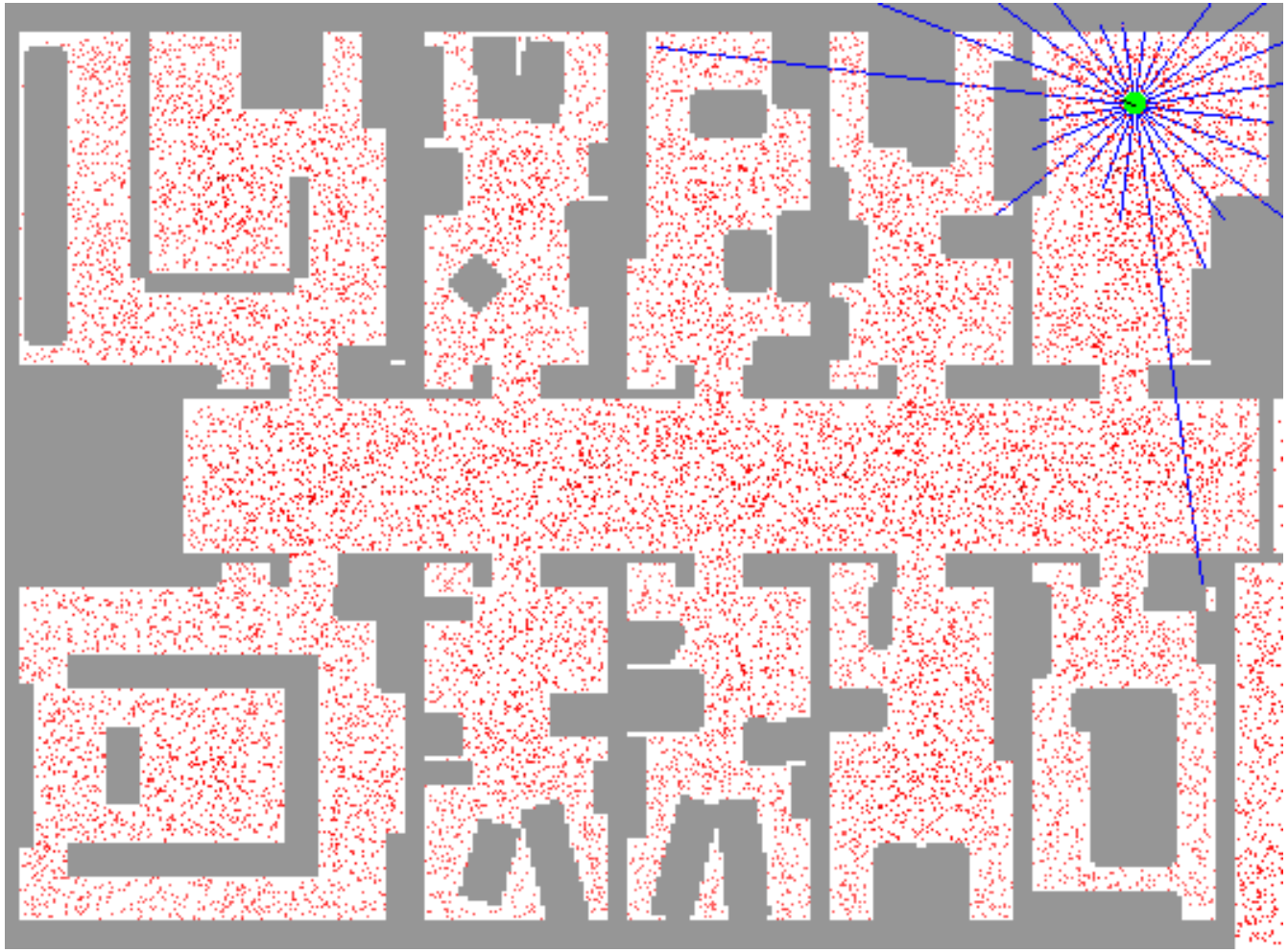
Beyond Gaussian Error Models

- We have seen through the Central Limit Theorem that given enough samples, each independent random variable is normal distributed.
- What if we can't *separate one from the other?*
- What if a single hypothesis is too limiting?



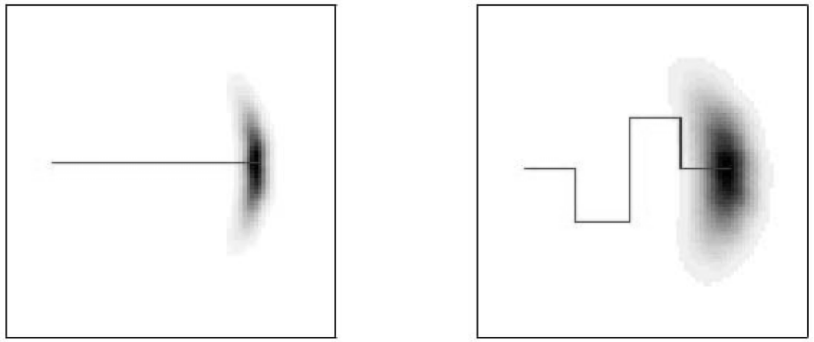
- Before the name "Particle Filters", the first stages were known as "Conditional Density Propagation", or "CONDENSATION".
- Fantastic section on Particle Filters can be found in Thrun's "Probabilistic Robotics", https://robots.stanford.edu/probabilistic-robotics/

Motivating example: Robot localization using maps and sonar



What's happening here?

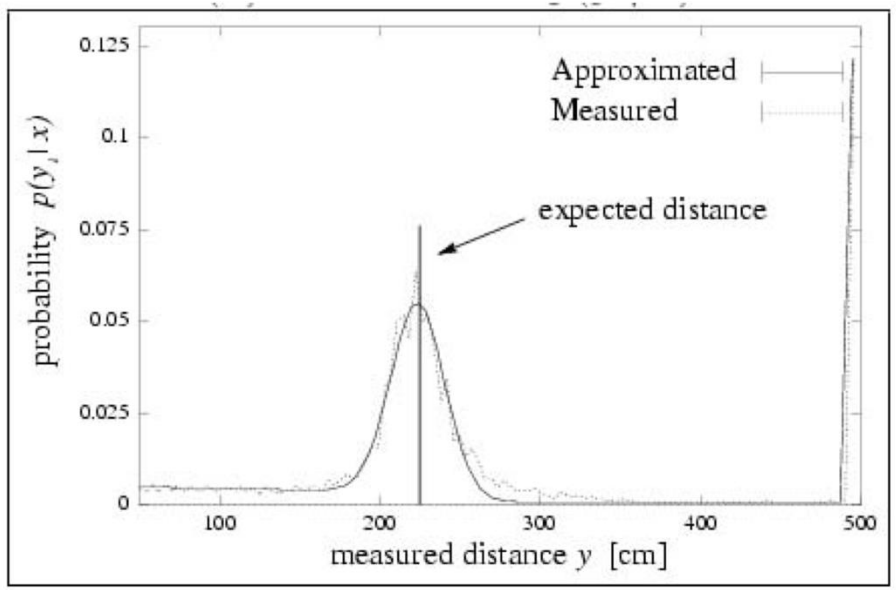
- Classical robot localization problem.
- The robot's state vector can be given as $\mathbf{x} = (x, y, \theta)^\top$.
- As we move it, its positional uncertainty increases with a density $p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{u}_{t-1})$.



- You can also think of a perception model

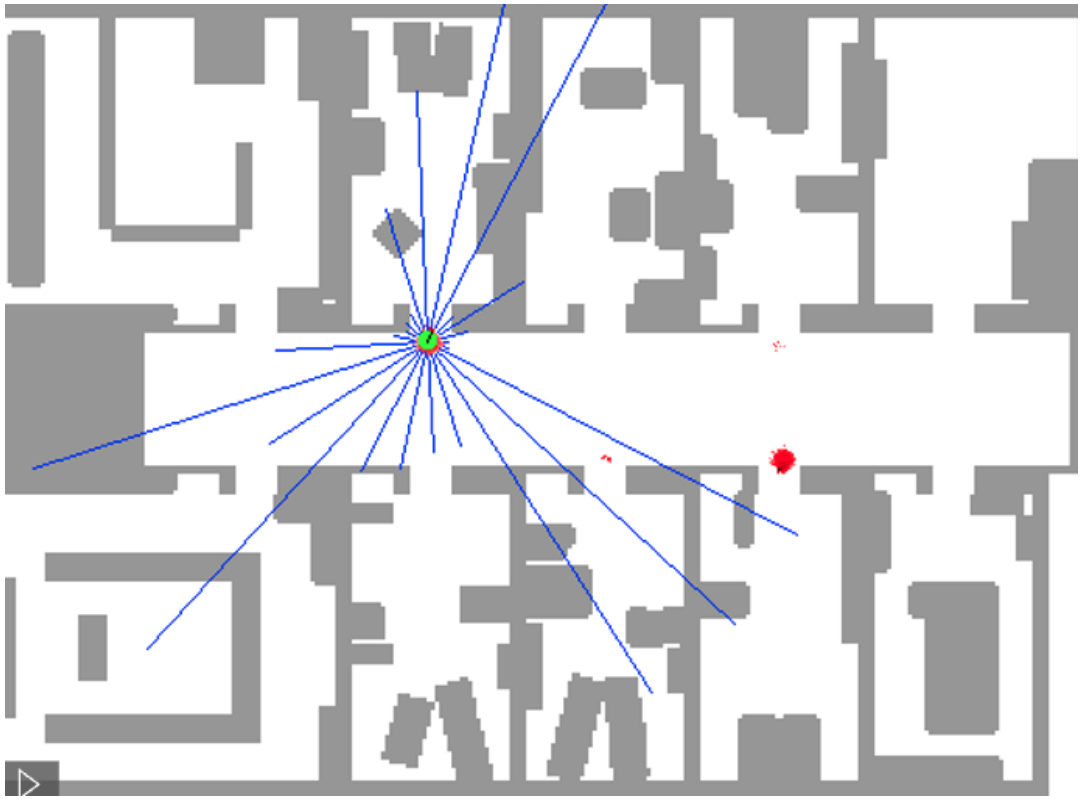
$p(\mathbf{z}_t | \mathbf{x}_t)$.





The Problem with Kalman Filters in Robot Localization

- Kalman Filters only represent state variables as single Gaussians.
- What if the robot can be in one of two places?



Particle Filters (a.k.a. Sequential Monte Carlo)



- Represents PDF as a set of samples (particles)
- Each particle contains one set of values for the state variables.
- Good for non-Gaussian, multi-modal PDFs.
- Find an *approximate* solution using a *complex* model (arbitrary pdf) rather than an *exact* solution using a *simplified* model (Gaussians).

How to find samples

- **We want to sample from the *posterior* distribution $p(\mathbf{x}_t|d_{0,\ldots,t})$ ($d_{0,\ldots,t}$ is "all the data from time 0 until now")**
- **Let's simply call it $p(\mathbf{x})$.**
- **But we don't have the explicit representation of the full PDF to synthesize samples.**
- **We *can* sample from the *prior* belief – let's call it $q(\mathbf{x})$.**
- **Update using observations:**
  - For each sample,
  - compare $p(\mathbf{x})$ to $q(\mathbf{x})$
  - and adjust by finding *importance weights.*



1. Sample from prior belief $q(\mathbf{x})$ (e.g. the uniform distribution)

distribution)

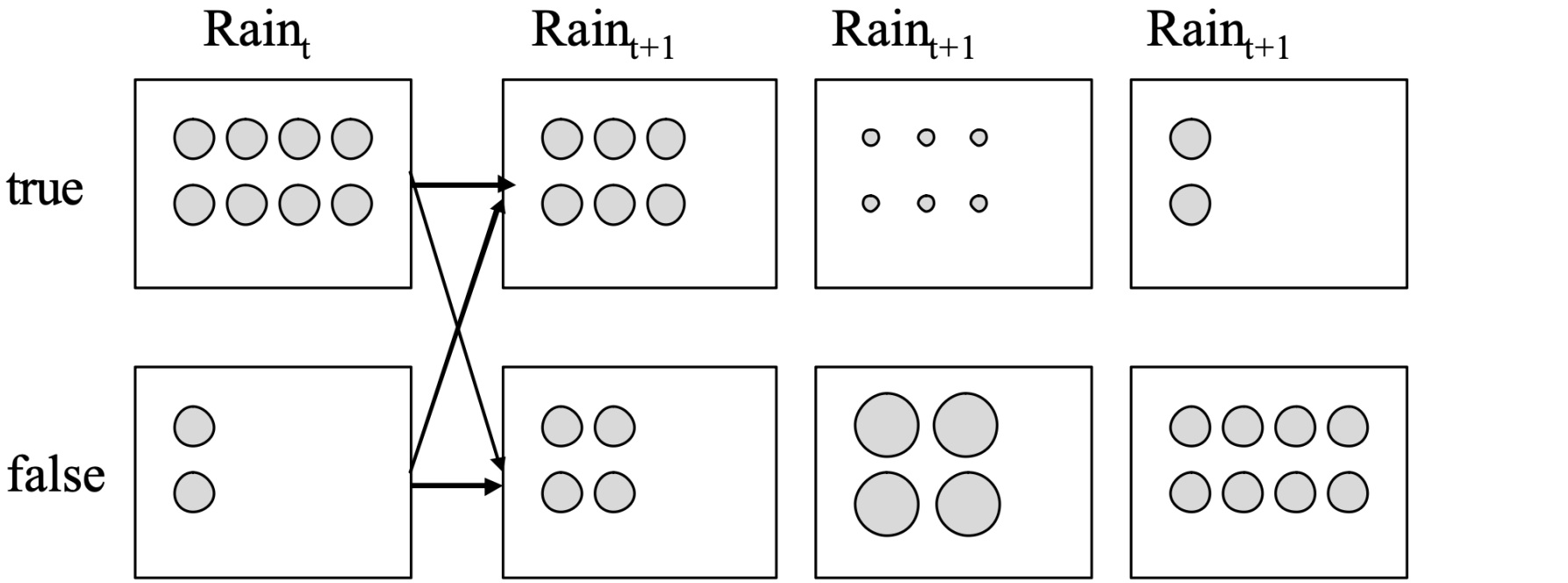


2. Compute importance weights, $w(\mathbf{x}) = p(\mathbf{x})/q(\mathbf{x})$



3. Resample particles according to importance weights to get $p(\mathbf{x})$.
Samples with high weights chosen many times, density reflects PDF.

Discrete Importance Resampling



Belief $p(\mathbf{x})$ from last time step

Prediction (sample from $q(\mathbf{x})$)

Update (weight samples by importance)

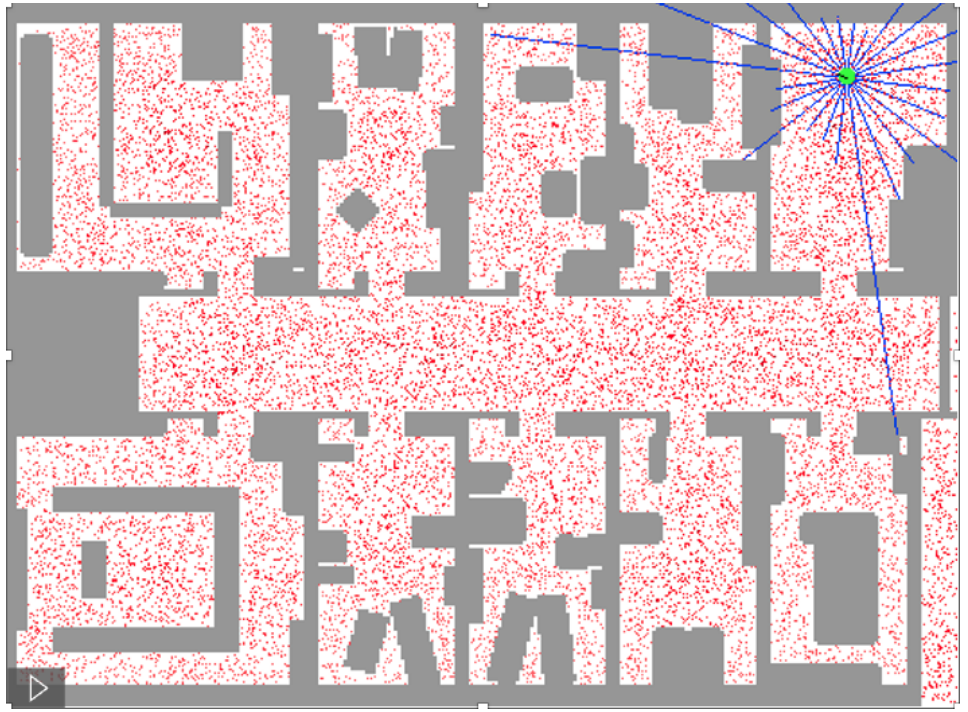
Resample according to weights to get $p(\mathbf{x})$ for new time step

Why Resample?

- **If you keep old particles around without resampling:**
  - Particle Depletion
  - Areas with high probability in posterior not represented well
  - Density of particles doesn’t represent pdf

Robot Localization – The Algorithm

- **Start: Sample with initial PDF,** $p(\mathbf{x}_0)$ **.**
  - For localization, no idea where the robot is ➔ throw particles everywhere!
- **For each time step, loop with three phases:**
  - Prediction
  - Update
  - Resample



Calculation of Belief for Robot Localization

Normalization Constant

PDF of current state | Perception Model | Motion Model | PDF of last state

$$p(\mathbf{x}_t|d_{0\ldots t}) = \eta \cdot p(\mathbf{z}_t|\mathbf{x}_t) \int p(\mathbf{x}_t|\mathbf{u}_{t-1}, \mathbf{x}_{t-1}) p(\mathbf{x}_{t-1}|d_{0\ldots t-1}) d\mathbf{x}_{t-1}$$

Resample

$p(\mathbf{x})$ - the posterior probability, our belief about the state variables

Update

$w(\mathbf{x})$ - the posterior probability, our belief about the state variables

Prediction

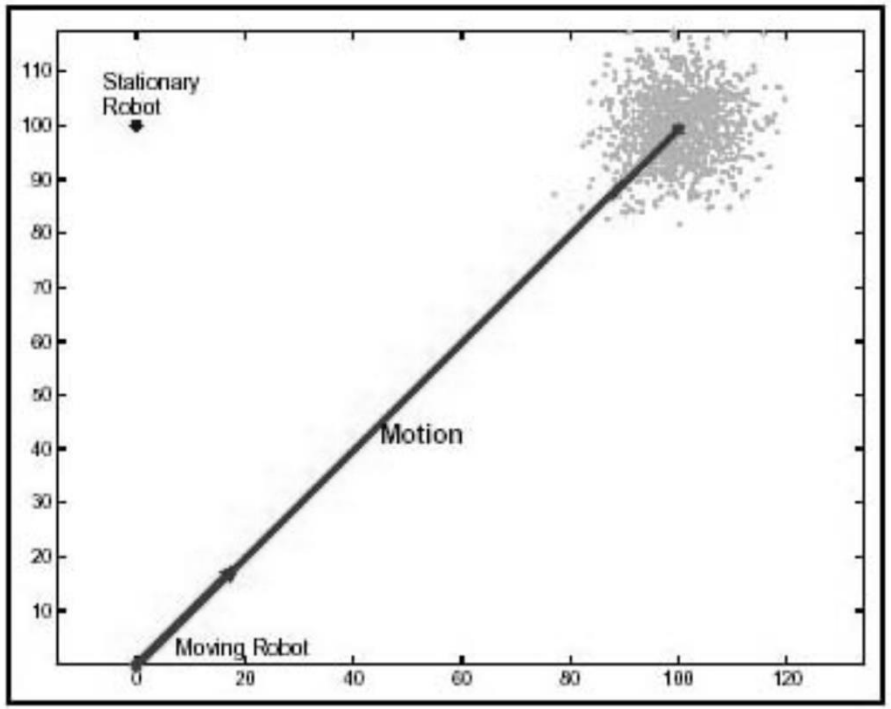
$q(\mathbf{x})$ - the prior probability

1. Prediction

- **Prediction: For each particle, sample and add random, noisy values from action model.**

- **Resulting proposal distribution**

$$q(\mathbf{x}) \approx p(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{u}_{t-1})p(\mathbf{x}_{t-1}|d_{0\ldots t-1})d\mathbf{x}_{t-1}$$



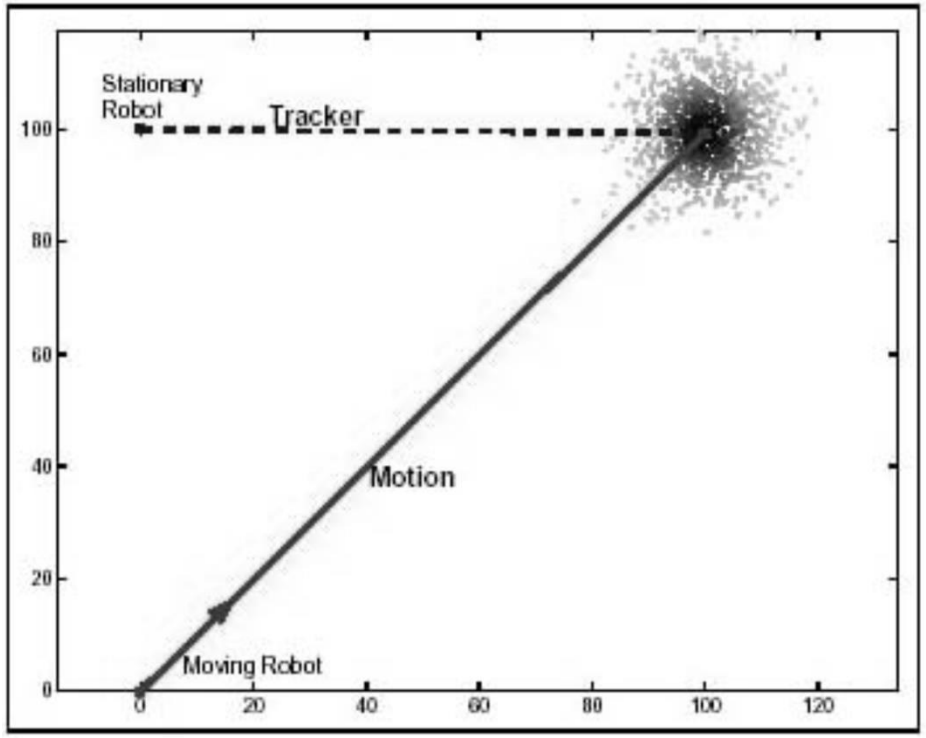
2. Update

- Each particle's weight is the likelihood of getting the sensor readings from that particle's hypothesis.

- The weight associated with each particle is

$$w(\mathbf{x}) = p(\mathbf{x})/q(\mathbf{x}) = p(\mathbf{z}_t|\mathbf{x}_t)$$

normalized so that all the weights sum to 1.



3. Resample

- New set of particles are chosen such that each particle survives in proportion to its weight.



- Resulting distribution is

$$\underbrace{p(\mathbf{x}_t|d_{0\ldots t})}_{\text{Posterior}} = \underbrace{\eta \cdot p(\mathbf{z}_t|\mathbf{x}_t)}_{\text{Weights}} \int \underbrace{p(\mathbf{x}_t|\mathbf{u}_{t-1}, \mathbf{x}_{t-1}) p(\mathbf{x}_{t-1}|d_{0\ldots t-1}) d\mathbf{x}_{t-1}}_{\text{Prior}}$$

Intro Summary

- **Particle filters represent pdfs using sample densities.**
- **New belief can be found using the prediction-update-resample loop.**
- **For certain applications, Particle filters can be better than Kalman filters.**

	Kalman Filter	Particle Filter
Optimal PDF	Gaussian	Arbitrary
Multimodal	No	Yes
Multi Hypothesis	No	Yes
Exactness	Exact	Approximate

Intro Summary

- **Note that there is no explicit state maintained, just a "cloud" of particles.**
- **You can obtain an estimate at a particular time by querying the current particle set.**
- **Some approaches:**
  - "Mean" particle
    - Weighted sum of particle
    - Confidence: inverse variance
  - Mode finder – model the mean of the tallest peak
- **Of course you can always transition:**
  - Start the robot localization problem with a Particle Filter
  - Once you've eliminated all except "one heap" of particles,
  - model the distribution you find there with a Kalman filter.

Caveats

- **Number of Particles is an important performance factor.**
  - Want as few particles as possible for efficiency.
  - But need to cover state space sufficiently well.

➔ **Worst-case complexity grows exponentially with state space dimensions.**

- **Multimodal densities possible, but still single object.**
  - Interactions between multiple objects require special treatment.
  - Not handled well in the particle filtering framework (state space explosion)

Visualization, and some more Examples









Literature / Links

1. **S.Thrun &al., *Probabilistic Robotics*, https://robots.stanford.edu/probabilistic-robotics/**



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Thank you very much!